

Data Collection and Preprocessing Phase

Date	20 th July 2026
Team ID	Akshat oberoii
Project Title	Food Ordering Behaviour Analysis
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	A comprehensive food ordering transactions dataset (~50,000 orders) covering six cities (Bangalore, Chandigarh, Delhi, Hyderabad, Mumbai, Pune), six cuisine types (Biryani, Chinese, Desserts, Fast Food, North Indian, South Indian), four meal types (Breakfast, Lunch, Dinner, Snacks), company/dining segments (Alone, Family, Friends, Partner), customer ratings, delivery time, order value, restaurant type, and customer age group, used to evaluate food ordering behaviour and consumer trends.
Data Cleaning	Missing Values: Isolated and removed incomplete order records to prevent skewed KPI totals. Duplicates: Dropped redundant Order IDs to guarantee accurate order counts. Error Correction: Trimmed double spaces, normalized inconsistent city and cuisine name casing, and stripped stray whitespace from categorical fields (City, Cuisine, Meal Type).
Data Transformation	Calculated Fields: Formed average rating and average delivery time calculated fields grouped by meal type and city. Aggregations & Pivots: Implemented SUM/COUNT calculations for Count of Order Id, and structured pivots for revenue contribution by meal type. Filtering: Excluded orders with unrealistic delivery time or order value outliers falling outside normal operational ranges.
Data Type Conversion	Validated that Order Id is classified as a non-numeric string identifier, Order Value and Avg. Rating fields are assigned floating-point decimals, and Time Taken to Order is mapped to a numeric (integer/bin) field to prevent aggregation errors inside Tableau.
Column Splitting and Merging	Merged related categorical fields (such as City and Restaurant Type) where applicable for combined breakdowns, and isolated multi-value strings (e.g., combined cuisine/meal-type tags) into clear, singular individual columns to enable responsive dashboard filtering.
Data Modeling	Established logical connections between the core orders table and lookup dimensions (City, Cuisine, Meal Type, Company Type) using Order Id as the index key. Calculated global operational metrics such as Total Orders KPI (50,000), delivery time distribution, and revenue contribution by meal type.
Save Processed Data	Exported the fully standardized dataset from raw system states into a permanently structured CSV file (food_ordering_behavior_dataset.csv), explicitly adjusting column formatting to ensure complete field visibility before connecting to Tableau Public for visualization.